Research Proposal
Section: C
Group: 102

Does GDP Truly Measure A Country's Growth?

Introduction and Literature Review

A. Identification of specific problem area

GDP or Gross Domestic Product is the monetary value of all the final goods and services produced within the borders of a country in a given period of time, usually a year or a quarter. GDP can be calculated by the production approach (adding up the value added in each industry), the income approach (adding up wages, profits, rents, and taxes), or the expenditure approach (C + I + G + NX, where C is consumption, I is investment, G is government spending, and NX is net exports). In theory all three should yield the same result.
Nominal GDP is based on current prices . Real GDP is adjusted for inflation and gives a better idea of the real growth in output over time .
GDP indicates the size and growth of an economy but it doesn't reflect income inequality, unpaid work, environmental costs or overall well-being – it's an activity measure not a happiness measure.
We use GDP (and GDP per capita) to compare economies in different countries and in time and to provide information for decisions on fiscal/monetary policy. We often add to it with other indicators (HDI, GNI, etc.) to provide a fuller picture.

Why this research is important & Motivation to choose this research:
This research is to understand whether GDP completely represent a country's condition, like what things does GDP show and what it hides.

- Informs policy-making by means of reliable research on GDP, allowing governments and central banks to formulate effective fiscal and monetary policies—such as tax rates, interest rates, and public spending—on the basis of the actual state of the economy.
- Provides a framework for understanding economic growth and fluctuations—by observing trends in GDP, we can identify periods of growth, recession or stagnation and react quickly to stabilize the economy.
- GDP data and the forecasts that accompany it are indispensable to investors, financial institutions and businesses in making their investment and business decisions. They are put to use to gauge market conditions, put capital where it is needed and chart a course for any expansion or contraction.
- Then there is the matter of development analysis and cross-country comparison. Research into GDP makes it possible to measure living standards

and economic output from one nation to another, in turn revealing the structural elements or policies at work behind the growth.
- With time, the methodology and measurement have been put to better use. Continuous research has served to refine the way GDP is computed, for instance by factoring in sustainability, digital goods and the informal economy; this lends the metric a greater degree of accuracy and pertinence to today's economic landscape.

B. Prevalence, scope of problem.

Issues/problems that this research addresses:

- There is the matter of income distribution, which GDP does not address. While it is a measure of total output, the figure gives no sense of how wealth and income are apportioned across the population. Inequality can be masked as a result, even in the face of an expanding economy.
- Excludes non-market and unpaid work – Household labor, caregiving, volunteer work and the informal/black economy contribute to well-being and society but are not captured in GDP figures.
- Ignores environmental and social costs – GDP counts economic activity as good even if it is due to resource depletion, pollution or activities that are damaging to the environment.
- A high GDP is no guarantee of quality of life or well-being. One will not find that it translates to superior health, education or happiness; a nation may boast a large GDP while its human development indicators tell a different story.
- Then there are the problems of measurement and the fact that the figure is open to manipulation. Unproductive outlays such as those for military action or in the wake of a disaster can put an artificial sheen on the numbers. And when it comes to making comparisons from one country to another, the task is made difficult by variances in size of the informal sector, exchange rates and the methodologies employed.

In this research, the scope of investigation is broader and GDP is compared with other indicators of development.

C. A summary of what is known (literature review)

GDP is one of the main measures employed to indicate a country's economic growth since it reflects the total value of the goods and services that are produced in that country and a higher GDP usually signifies that the economy is growing.
Earlier research has, however, demonstrated that GDP alone is incapable of measuring a country's total growth; although GDP focuses on economic production, it does not fully capture the way people actually live.

Stiglitz, Sen and Fitoussi (2009) stated that although GDP is useful as a means of measuring economic performance it does have some important limitations. For instance, GDP does not indicate how income is distributed among individuals. A country might have a high GDP yet a large number of people could still be poor. Also, GDP does not adequately take into account factors such as health, education and quality of life.
The Human Development Index (HDI) is also a significant indicator of development since it takes into account health, education and standard of living as well as income. As a result, the HDI provides a more comprehensive picture of development than GDP does.
The OECD also looked into the concept of "Beyond GDP". Its research indicated that GDP fails to consider important factors such as income inequality, environmental conditions and people's well-being. As a result, relying solely on GDP can lead to an incomplete understanding of a country's level of development.
The World Bank has also stated that GDP does not clearly indicate whether economic growth is sustainable. For instance, a country might raise its GDP by using a large quantity of its natural resources. This could result in growth in the short term, but it might later cause environmental problems.

D. Overview of what we don't know

Earlier research has clearly demonstrated that GDP can be used as a measure of economic growth, but it does not give a full picture of a country's development; there is therefore still a need to compare GDP with other indicators such as the HDI in order to ascertain whether greater economic growth also implies better living conditions.

E. Statement of purpose and research question(s)

The first aim of this study is to analyze GDP as a measure of the economic growth of a country. The examination will be focused on explaining what GDP tells us about the economic state of a country and what information it provides. This examination will help in determining the value of GDP before comparing it with other measures.
The second aim of this study will be to conduct a comparison between GDP and Human Development Index (HDI) and other indicators for development. It must be noted that HDI covers aspects like health, education and quality of life, which cannot be quantified using the measure of GDP. Through comparison of these development indicators and GDP, the study intends to determine if better economic growth results in better development indicators.
Another aim is to carry out an analysis of the changes and trends in the GDP and in a number of development indicators over a specific period, in order to ascertain whether fluctuations in economic growth have an effect on human development, since this comprehensive approach is being adopted.

The fourth aim is to find out whether the GDP measure alone is sufficient enough to represent the growth and development of the nation. The research neither takes it for granted nor denies the idea that GDP measure is either absolutely sufficient or absolutely insufficient. On the contrary, the conclusions will be made on the basis of comparative analysis of selected indicators.
In general, the research aims at clarifying the level to which GDP is able to describe the growth of the country and its shortcomings. Moreover, it shows that the use of several indicators can result in a more thorough description of the country's development. These results help determine whether the economic performance and the improvement of living standards always go together.

Methodology

A. Design

A mixed research design has been employed in this study. The reason for this is that it is only by combining figures with actual stories that we can obtain a comprehensive understanding of growth. The particular design consists of a comparative data analysis together with case studies. The reason for choosing this approach is that it is necessary to examine long-term trends as well as to see how real people actually live. The specific research methods involve the use of groups of different nations. Surveys or control groups will not be used. The main reason for this is that the study is concerned with national level data rather than with testing individual people in a laboratory setting. By comparing rich and poor countries it will be possible to determine whether Gross Domestic Product operates in the same way everywhere in the world. This simple design is both strong and clear and will help us to arrive at the exactly correct answer to our primary research question without becoming lost in an excessive number of additional details today.

B. Sample and Procedures

The primary group in our study consists of all the countries which collect economic and social data. The sample we have chosen is a particular set of twenty countries; ten developed ones and ten developing ones. The expected number will be exactly twenty nations. The demographics will be fully balanced in order to give an impartial view. We desire a good mixture of rich and poor countries. This mixture will ensure that the final results of our study are honest and genuinely reflect how the real world is working at the present time.
The data, which can be obtained for free from reliable sources, will be used to select the sample. Once this has been done, the next step will be to collect the scores that these entities obtained from the year 2000 up to 2020. All of this data will then be entered into clear tables in order that a detailed comparison of the figures over a twenty year period may be made. The fact that the time span in question is so long

allows us to gain a reliable and consistent insight into how a country develops over time.

C. Measurement

In the study we carried out, four standard indicators—those which are widely used in many countries—were employed in order to assess national progress beyond just monetary value.

- Real GDP per capita: This figure can be obtained from the World Bank and is used for measuring a country's economic output per person without the distorting effect of inflation [1].
- Human Development Index (HDI): a tool developed by the United Nations to measure a country's overall social and economic progress [2].
- Gini Coefficient: A statistical measure used to determine the economic inequality, such as the distribution of wealth within a country [3].
- CO2 Emission: Its data is given by the World Bank, and it can be used to know the progress of a country, too [4].

D. Ethical or Political Issues

Since the data for this study are taken from public sources and verified by global institutions, the risk of privacy issues or violations of privacy rights is reduced to zero. To reduce political bias or selective reporting, which can distort research, we collected data from neutral international sources such as the United Nations and the World Bank, which are not aligned with any particular government.

E. Analysis Plan

For Data analysis, we used 4 structural computation and statistical steps:

1. We had to normalise the data since the different units of our data and the indicators would have meant that their use was not appropriate and errors could have occurred; min-max normalisation is the best method of normalising the data.
2. Correlation: It is used to see the correlation between our indicators and GDP that how changes in one of the indicators can affect GDP if it affects it in any way. It can use Spearman correlation here.
3. Comparison of rankings: Because we have different indicators and the countries' ranks vary greatly, we calculate how much they differ by taking the rank for GDP and subtracting the rank obtained on the other indicators.
4. The environmental review examines whether the country is achieving a rise in its GDP without increasing its CO2 emissions and whether this kind of growth is sustainable, or whether its GDP is merely the result of environmental degradation.

F. Limitations

There are some limitations of the methods that we have used. Some of the limitations are:

- We have not included informal transactions in accounts, since GDP uses only formal transactions; therefore, there is no way to obtain informal data, which can affect the results.
- Although National GDP figures are regularly updated, inequality indices such as the Gini index are updated only at intervals and are based on data from national household surveys, which may eventually lead to a gap in our research.
- Measures such as the HDI usually give all the different factors fixed mathematical weights, and thus are unable to fully take into account the circumstances of a country. For instance, a country might have the best education system but the worst healthcare, yet the HDI or similar measures would rate it as average, which is not in accordance with the actual situation.

IV. References

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16. World Bank, "CO2 emissions (metric tons per capita)," World Development Indicators, 2024. [Online]. Available: https://data.worldbank.org/indicator/EN.GHG.CO2.PC.CE.AR5

Additional source links for ideas/concepts used in this proposal (added to support proper attribution):

1. Stiglitz, J. E., Sen, A., & Fitoussi, J.-P. (2009). Report by the Commission on the Measurement of Economic Performance and Social Progress — full report (basis for the discussion of GDP's limitations regarding income distribution, health, education, and quality of life). [Online]. Available: https://ec.europa.eu/eurostat/documents/8131721/8131772/Stiglitz-Sen-Fitoussi-Commission-report.pdf
2. OECD, Beyond GDP: Measuring What Counts for Economic and Social Performance (2018), by J. Stiglitz, J. Fitoussi and M. Durand — source for the "Beyond GDP" discussion of inequality, environmental conditions and well-being not captured by GDP. [Online]. Available: https://www.oecd-ilibrary.org/economics/beyond-gdp_9789264307292-en
3. World Bank, The Changing Wealth of Nations — source for the discussion of GDP not capturing the sustainability of growth (resource depletion, environmental costs). [Online]. Available: https://www.worldbank.org/en/publication/changing-wealth-of-nations/report
4. Wikipedia, "Gross domestic product" — background source for the definition of GDP and the production, income, and expenditure approaches, and nominal vs. real GDP. [Online]. Available: https://en.wikipedia.org/wiki/Gross_domestic_product
5. UNDP, "Human Development Index," Human Development Reports Data Center — source for the definition and conceptual basis of HDI as a measure

combining health, education, and standard of living. [Online]. Available: https://hdr.undp.org/data-center/human-development-index
6. World Bank, "Gini index," World Development Indicators — source for the definition of the Gini coefficient as a measure of income/consumption inequality. [Online]. Available: https://data.worldbank.org/indicator/SI.POV.GINI
7. Wikipedia, "Feature scaling" — background source for the min-max normalization method referenced in the Analysis Plan. [Online]. Available: https://en.wikipedia.org/wiki/Feature_scaling
8. Wikipedia, "Spearman's rank correlation coefficient" — background source for the Spearman correlation method referenced in the Analysis Plan. [Online]. Available: https://en.wikipedia.org/wiki/Spearman%27s_rank_correlation_coefficient

Appendices

There is no need for us to get informed consent for this project since we shall carefully maintain records of our activities. We are not communicating with any human participants; rather, we are using public, historical data from earlier global reports. No harm will result from this. We adhere to all the privacy rules.
From an ethical point of view, there are no risks to human subjects since no humans are taken part in the research project that we are currently carrying out. However, we will make sure to report all the data in a completely honest manner without altering any of the figures to match our own beliefs. We will certainly make use of proper measures to protect the full truth of our research. It is of great importance to remain strictly neutral at all times. We will by no means allow our personal feelings to influence the final mathematical results in any way whatsoever.